

fumes. By 1903, an experimental electric railcar running on a military line in Germany set a new world speed record. The early 20th century also saw the appearance of the compression ignition, oil-fuelled locomotive – a precursor of the mass move to diesel traction that followed later.



▲ **Union Pacific Challenger No. 3999** and Union Pacific Big Boy no. 4011 double-heading a train on Sherman Hill, US, in 1946.

World goes to war

The railways proved to be invaluable during the American Civil War (1861–65) and played a crucial role once again, during the Great War as World War I (1914–18) came to be called. During the course of the four years, trains moved men, supplies, and munitions across great distances. Railway-mounted artillery was a common feature especially along the frontlines with France becoming the first country to use rail-mounted howitzers adapted from naval guns. Special locomotives were built in some countries and narrow-gauge railways were laid specifically, often close to frontlines, to serve the war effort.

The end of the war saw the creation of many new countries, which inherited these rail systems, adapting and reorganizing them to suit their